



Hydrochemical Responses to Land Use and Seasonality in a Semiarid Watershed Under Anthropogenic Pressure

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Abstract Anthropogenic activities and hydroclimatic variability have intensified hydrochemical changes and water-quality degradation in semiarid watersheds, increasing the risks of salinization and eutrophication. This study evaluated the influence of land use and seasonal variability on the hydrochemical and microbiological characteristics of surface waters in the lower Piranhas-Açu River basin, northeastern Brazil. Water samples were collected at seven monitoring points downstream of the Armando Ribeiro Gonçalves Reservoir during dry and rainy periods between 2019 and 2021. Physical, chemical, and microbiological parameters, including pH, electrical conductivity (EC), Na^+ , K^+ , Ca^{2+} , Mg^{2+} , Cl^- ,

CO_3^{2-} , HCO_3^- , alkalinity, nitrogen, phosphorus, thermotolerant coliforms, and chlorophyll-a, were evaluated using Pearson correlation, principal component analysis, cluster analysis, and factor analysis. Multivariate analyses identified three distinct water-quality groups, while factor analysis retained three factors explaining 71.93% of the total variance and representing trophic conditions, ionic enrichment, and carbonate–bicarbonate dynamics. Dry periods were associated with greater salinity, alkalinity, phosphorus concentrations, and phytoplankton response, whereas rainy periods showed higher N, K^+ , and HCO_3^- levels and lower overall mineralization. Urbanized areas were mainly associated with phosphorus enrichment

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and eutrophic indicators, while agricultural activities and shrimp farming coincided with higher dissolved-ion concentrations, EC, and alkalinity. Overall, land-use pressures and hydroclimatic seasonality increased spatial and temporal water-quality heterogeneity, supporting multivariate approaches for integrated water-resource management in semiarid regions.

Highlights • Multivariate analyses revealed distinct spatial and temporal water-quality patterns.

- Factor analysis identified three hydrochemical factors explaining 71.93% of total variance.
- Rainfall reduced P and chlorophyll-a by 95.10% and 78.90%, respectively.
- Dry-season waters showed greater ionic enrichment and alkalinity than rainy-season waters.
- Contrasting land uses coincided with distinct hydrochemical and nutrient-related spatial patterns.

Keywords Water resources · Anthropization · Seasonality · Semiarid

1 Introduction

The Brazilian semiarid region is characterized by marked spatial and temporal irregularity in rainfall, shallow and often stony soils, and vegetation dominated by the Caatinga biome. Under these conditions, reservoirs play a strategic role in ensuring water availability, food security, and socioeconomic development. However, increasing anthropogenic pressures have progressively compromised both the quantity and quality of available water resources. Domestic sewage, industrial effluents, agricultural activities, and intensive water use contribute to the degradation of surface and groundwater, while excessive abstraction may further affect aquifer recharge and long-term water availability (Silva et al., 2024).

Water quantity and quality are also strongly regulated by hydroclimatic processes that control stream-flow, pollutant transport, dilution, and concentration within aquatic systems (Mesquita et al., 2020; Wiegand et al., 2021). Variations in rainfall and hydrological regimes can modify pollutant loads and concentrations (Couture et al., 2018; Me et al., 2018; Messina et al., 2020), while the increasing frequency and intensity of droughts and floods impose

additional challenges for water-resource management (Shrestha et al., 2017). These effects are particularly pronounced in semiarid environments, where prolonged droughts reduce stored water volumes and enhance evapoconcentration (Rocha Junior et al., 2018), whereas intense rainfall following dry periods can increase surface runoff and rapidly transport nutrients and other constituents from the watershed into water bodies (Mesquita et al., 2020; Santos et al., 2014).

In addition to hydroclimatic variability, land use and watershed characteristics strongly influence limnological and hydrochemical conditions (Me et al., 2018; Menezes et al., 2016; Soares et al., 2012). Urbanization can increase nutrient and microbial inputs through untreated or insufficiently treated wastewater, whereas agricultural and livestock activities may contribute fertilizers, agrochemicals, animal wastes, and dissolved salts through surface runoff and irrigation return flows (Araújo et al., 2019; du Plessis et al., 2015; Me et al., 2018). In semiarid regions, these pressures interact with high evaporation rates and seasonal fluctuations in water volume, favoring the concentration of salts and nutrients and increasing the susceptibility of aquatic ecosystems to salinization and eutrophication (Macedo & Tavares, 2010; Manaouch et al., 2020; Rocha Junior et al., 2018). Consequently, systematic water-quality monitoring is essential for detecting changes associated with land use and land cover and for supporting effective planning and management of water resources (Carvalho et al., 2022).

Within this regional context, the Armando Ribeiro Gonçalves Reservoir stands out as the largest reservoir in the state of Rio Grande do Norte and plays a strategic role in water supply, maintenance of perennial river flow, flood control, irrigated agriculture, aquaculture, industrial activities, and tourism. Despite its socioeconomic and environmental importance, studies conducted since 2000 have consistently reported eutrophic conditions, recurrent dominance of potentially toxic cyanobacteria, and marked temporal variability in phytoplankton communities, particularly in response to seasonal and hydrological changes (Câmara et al., 2015; Chellappa et al., 2009; Costa et al., 2006; Vieira et al., 2015). Costa et al. (2006) documented high cyanobacterial dominance, including *Microcystis* spp., *Cylindrospermopsis raciborskii*, and *Aphanizomenon* spp., together

with the occurrence of microcystins and saxitoxins. Chellappa et al. (2009) subsequently showed that seasonal alternation modified phytoplankton composition, with *Planktothrix agardhii* prevailing during the dry season and *Microcystis aeruginosa* during the rainy season. Under an anomalous rainfall regime, Câmara et al. (2015) observed that increased water levels, water-column mixing, and nutrient pulses reduced the dominance of potentially toxic cyanobacteria and increased phytoplankton diversity. Similarly, Vieira et al. (2015), in a longer-term assessment from 2009 to 2011, reported pronounced temporal changes characterized by the succession of cyanobacterial species and the persistence of eutrophic conditions. Collectively, these findings indicate that water quality in the Armando Ribeiro Gonçalves Reservoir has historically reflected the interaction between nutrient enrichment, phytoplankton dynamics, and hydroclimatic variability.

Nevertheless, direct comparisons among studies conducted in the region require caution. Differences in the geographic location of sampling sites expose water bodies to contrasting land-use, land-cover, and geogenic conditions, which may generate distinct hydrochemical, trophic, and microbiological responses. Moreover, the irregular distribution of rainfall modifies streamflow, runoff, and the concentration or dilution of salts and nutrients, while differences in sampling periods, analytical procedures, and seasonal conditions may further limit direct comparisons. Even with these constraints, previous studies provide an essential basis for understanding the temporal evolution of water quality and for identifying the environmental processes that may control its spatial variability (Câmara et al., 2015; Chellappa et al., 2009; Costa et al., 2006; Vieira et al., 2015).

Given this complexity, multivariate statistical approaches provide a robust framework for integrating multiple dimensions of water-quality variability. Pearson correlation analysis identifies associations among measured variables, hierarchical cluster analysis groups sampling sites or periods according to their overall similarity, and principal component analysis reduces dimensionality while revealing the dominant gradients structuring the dataset. Factor analysis complements these approaches by identifying groups of variables that exhibit similar patterns of variation and by supporting the interpretation of hydrochemical and environmental processes potentially associated with

those patterns (Hair Junior et al., 2014; Maia et al., 2019; Muniz et al., 2019). Together, these techniques allow spatial and temporal variability to be evaluated in an integrated manner and facilitate the interpretation of complex interactions among hydroclimatic conditions, land use, and water quality.

Although previous investigations of the Armando Ribeiro Gonçalves Reservoir have provided important information on eutrophication, phytoplankton dynamics, cyanobacteria, cyanotoxins, and selected physicochemical and microbiological indicators, integrated assessments linking spatial and temporal changes in downstream water quality with land use, land cover, and hydroclimatic seasonality remain comparatively limited. Addressing this gap is particularly important because water-quality assessment provides a fundamental basis for identifying environmental pressures, evaluating their potential impacts, and supporting mitigation and management strategies (da Silva et al., 2026).

Therefore, this study aimed to evaluate whether land use and land cover downstream of the Armando Ribeiro Gonçalves Dam are associated with spatial and temporal variations in the physical, chemical, and microbiological quality of water in the Piranhas-Açu River basin. It was hypothesized that spatial and temporal variations in water quality are jointly shaped by land-use pressures and hydroclimatic seasonality, leading to distinct hydrochemical conditions expressed through differences in salinity, nutrient dynamics, eutrophic responses, and microbiological contamination.

2 Material and Methods

2.1 Characterization of the Study Area

The research was carried out in the lower part of the Piranhas-Açu River basin, located in the Vale do Açu micro-region, occupying an area of 4756.1 km² which corresponds to 9.06% of the area of the state of Rio Grande do Norte, Brazil. Seven water collection points were established, representative of the predominant uses and occupation of the lower part of the basin, located downstream of the Armando Ribeiro Gonçalves Dam (ARGD) with a storage capacity of 2.4 billion cubic meters (IBGE, 2010), as shown in Fig. 1.

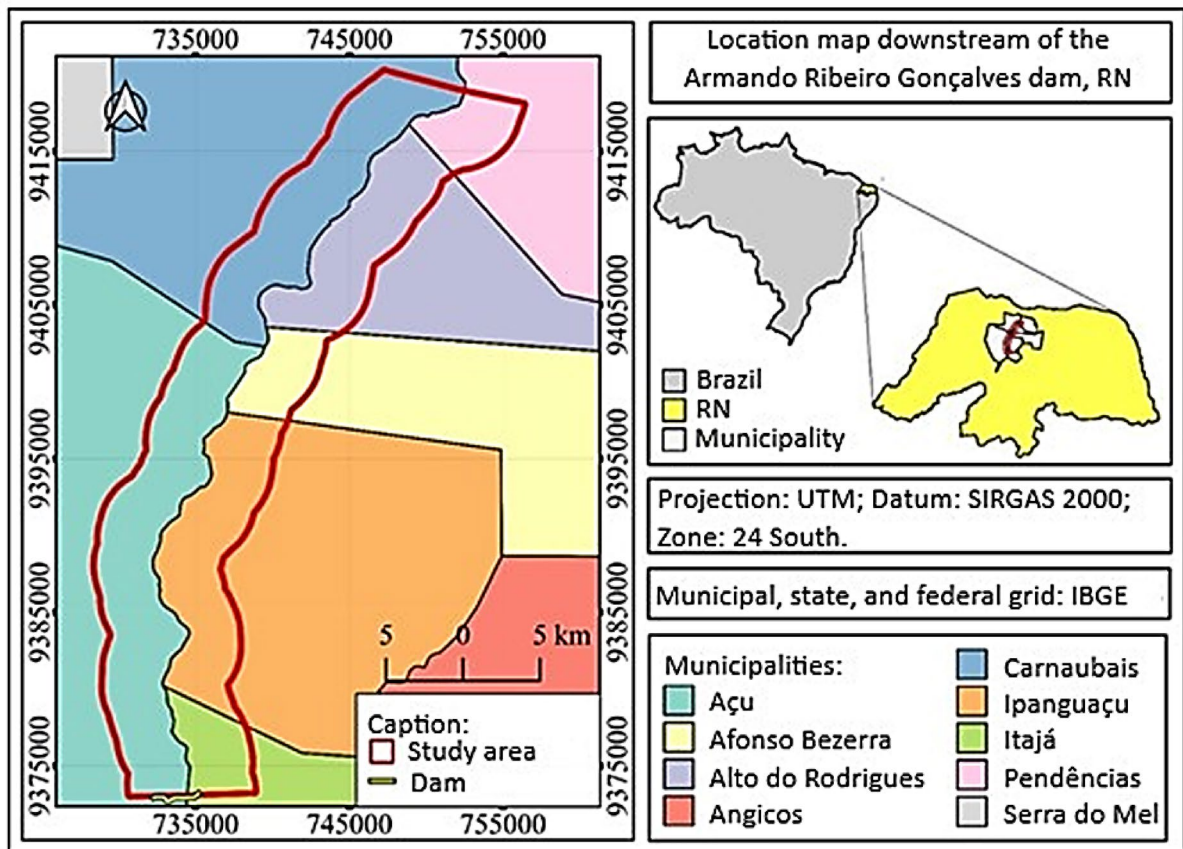


Fig. 1 Location of the study area downstream of the Armando Ribeiro Gonçalves Dam, Rio Grande do Norte state, Brazil

Geologically, the study area is located within the Borborema Province and is characterized by a heterogeneous assemblage of geological units mapped by the CPRM. These units include formations associated with coastal tablelands, alluvial and colluvial deposits, interbedded silt–clay and quartz-rich sandy sediments, undifferentiated migmatites, and quartz-rich sandy and conglomeratic sediments interbedded with silt–clay materials. This geological diversity contributes to spatial differences in sediment composition and may influence water–rock and water–sediment interactions along the study area (CPRM, 2005a, 2005b, 2005c, 2005d, 2005e), as shown in Fig. 2.

The soil classes identified in the study area, according to the Brazilian Soil Classification System, are Neossolo Flúvico, Neossolo Litólico, Cambissolo Háplico, and Latossolo Vermelho-Amarelo (Fig. 3). These classes broadly correspond, under

USDA Soil Taxonomy, to Fluvents and Orthents within the Entisols, Inceptisols, and Oxisols, respectively (Santos et al., 2025). Neossolo Flúvico is the predominant soil class, occurring mainly along the fluvial corridor, whereas Cambissolo Háplico, Neossolo Litólico, and Latossolo Vermelho-Amarelo occur in more restricted portions of the study area. (Santos et al., 2025).

According to the Köppen classification (Köppen, 1918), the study area has a BSh climate, with summer and fall rains and an average temperature of 26.5 °C (Alvares et al., 2014). The rainy season begins in February and lasts until May, with the wettest months being March and April, with an average annual rainfall of 654 mm. The average annual temperature is 28 °C and the relative humidity is 70% (INMET, 2025). The predominant vegetation is of the type Caatinga Hiperxerófila, Carnaubal and Halófitas.

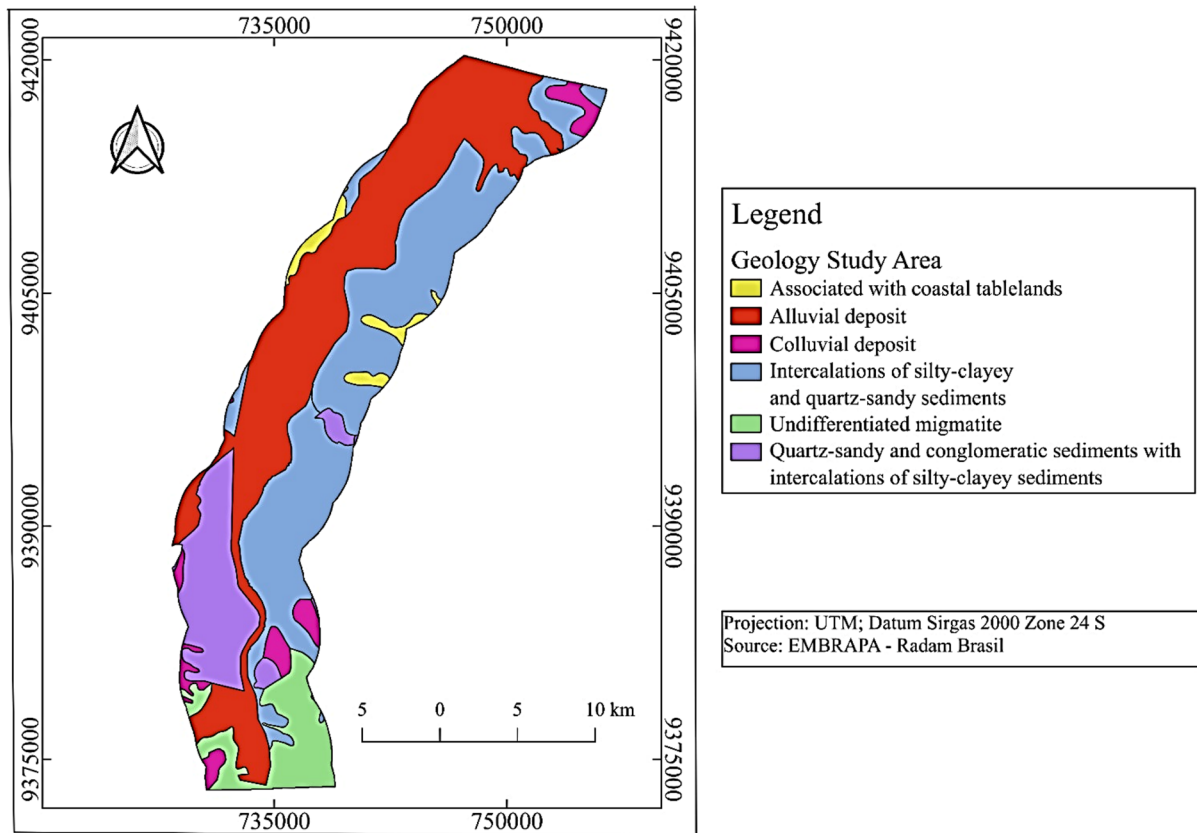


Fig. 2 Geological classes in the reach downstream of the Armando Ribeiro Gonçalves Dam, in the state of Rio Grande do Norte, Brazil

2.2 Land Use Diagnosis

Land use was determined by defining the use and occupation of the area downstream of the dam, showing areas of forest, non-forest natural formation, agriculture, non-vegetated areas and bodies of water (Fig. 4), within a 4-km buffer extending from each margin of the main channel of the Piranhas-Açu River. The Landsat satellite images, with a spatial resolution of 30 m, were taken from MapBiomas, collection 7.0 of 2021, and the map was then constructed using QGIS software, version 3.16 (MapBiomas, 2021).

2.3 Sampling and Rainfall Seasonality

Water samples were collected in 2019, 2020, and 2021 at seven sites located downstream of the dam (Fig. 3). The sampling sites were selected to represent predominant land-use and land-cover conditions

potentially influencing surface-water quality, while considering accessibility and field-sampling safety. The geographic coordinates of each site were recorded using a handheld Global Positioning System (GPS) receiver (Garmin GPSMAP 78 s, Garmin Ltd., Xizhi District, New Taipei City, Taiwan). Sampling campaigns were conducted under contrasting hydroclimatic conditions, including drier and wetter periods, to capture seasonal changes in the magnitude of water-quality parameters. This design allowed the assessment of processes such as evapoconcentration during drier conditions and dilution, hydrological renewal, and enhanced surface runoff during wetter periods, thereby improving the interpretation of the combined influence of geogenic characteristics and anthropogenic pressures across the study area (Fig. 5).

In 2019, samples were taken in June (C1), September (C2) and November (C3), the months

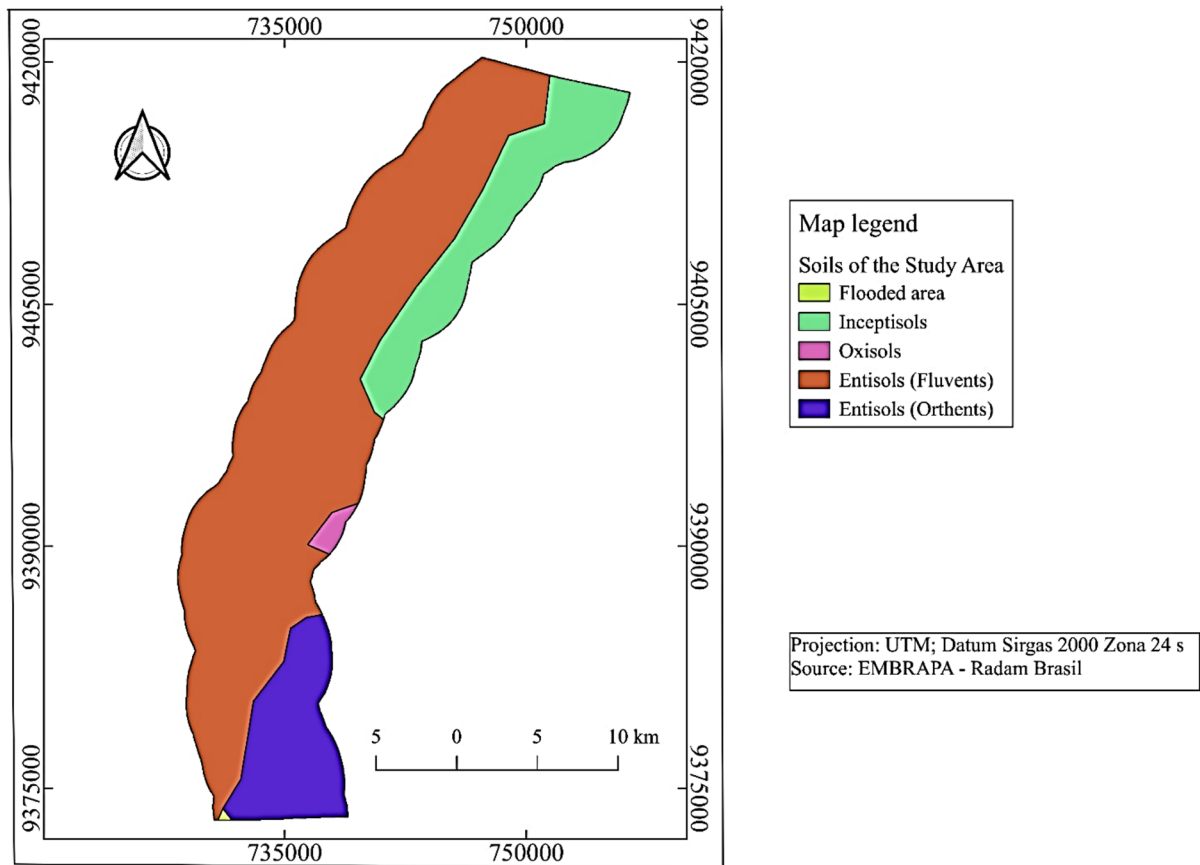


Fig. 3 Soil classes in the reach downstream of the Armando Ribeiro Gonçalves Dam, in the state of Rio Grande do Norte, Brazil

corresponding to the dry season in the region. For 2020 (C4) and 2021 (C5), collections were made in March, the months corresponding to the rainy season, as shown in Fig. 6.

2.4 Physico-chemical and Microbiological Water Analysis

To preserve sample integrity, the vials were stored in ice-cooled thermal containers, following the procedures recommended by APHA (2017), and subsequently transported to two laboratories at the Universidade Federal Rural do Semi-Árido (UFERSA), Mossoró Campus: The Soil, Water and Plant Analysis Laboratory (LASAP) and the Semi-Arid Limnology and Water Quality Laboratory (LIMNOAQUA). For clarity, the procedures adopted for sample collection, preservation, storage, and transportation. Water samples were collected from the surface layer (0–30 cm)

and stored in polypropylene bottles according to the requirements of each analytical procedure. Samples collected in 1 L bottles were used for the determination of pH, electrical conductivity (EC), major ions, nitrogen, and alkalinity. The pH and EC were measured immediately after the samples arrived at the laboratory using a benchtop pH meter and conductivity meter, respectively. Sodium (Na^+) and potassium (K^+) were determined by flame photometry; calcium (Ca^{2+}) and magnesium (Mg^{2+}) by EDTA complexometric titration using Eriochrome Black T as indicator; chloride (Cl^-) by titration with silver nitrate using potassium chromate as indicator; and carbonate (CO_3^{2-}) and bicarbonate (HCO_3^-) by acid titration using phenolphthalein and methyl orange as indicators, respectively, following Embrapa in triplicates (Silva, 2009) (Table 1). Total nitrogen (N) was determined by the Kjeldahl method, including digestion, distillation, and titrimetric quantification by Standard Methods

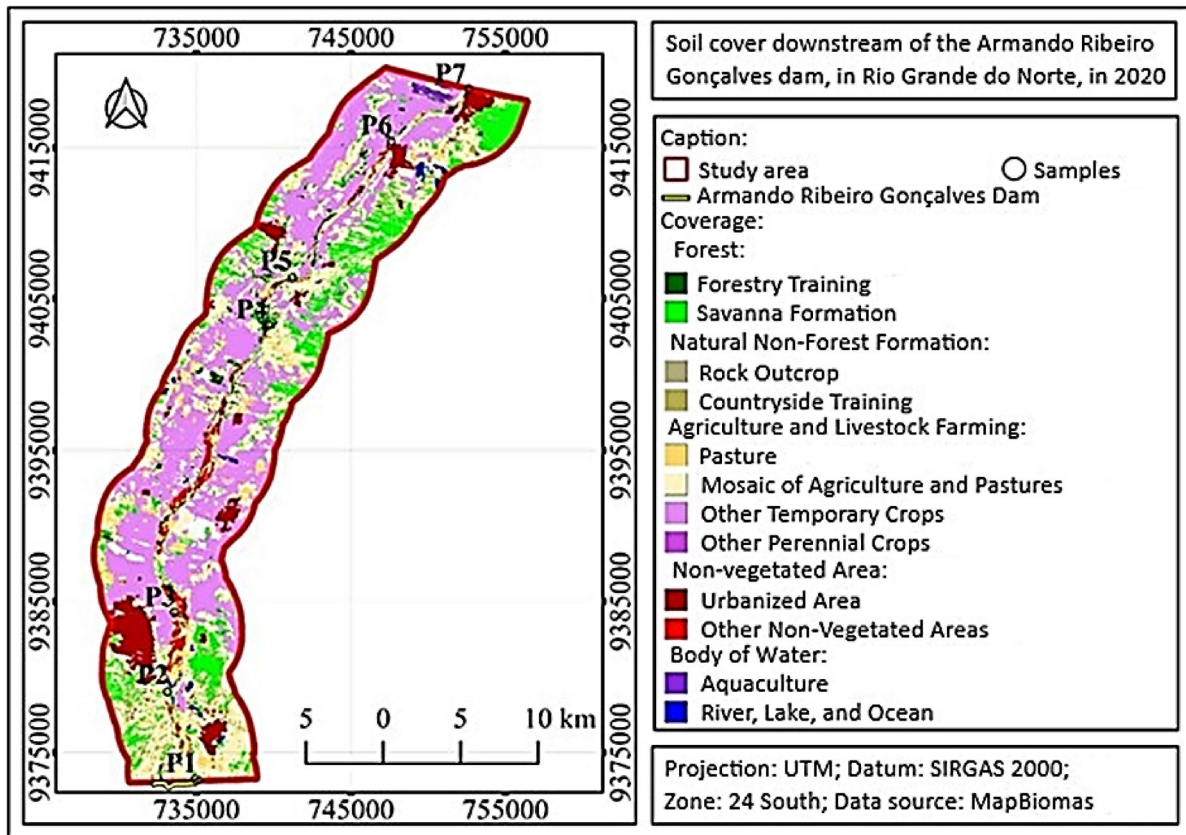


Fig. 4 Land use and occupation in the stretch downstream of the Armando Ribeiro Gonçalves Dam, in the state of Rio Grande do Norte, Brazil, and the respective collection points

4500-NH₃ B and 4500-NH₃ C. Alkalinity (ALK) was determined by volumetric titration with a standardized strong acid solution and expressed as mg CaCO₃ L⁻¹ by Method 2320 B of Standard Methods. Total phosphorus (P), collected in 300 mL bottles, was determined in unfiltered samples after persulfate digestion followed by the ascorbic acid colorimetric method and spectrophotometric quantification by Standard Methods 4500 P. Thermotolerant coliforms (TC), collected in 60-mL bottles, were quantified using the multiple-tube fermentation technique and expressed as MPN 100 mL⁻¹ by Standard Methods 9221 E. For chlorophyll a (Chl a), samples were collected in 1-L bottles protected from light with black tape, filtered, and subjected to pigment extraction with an organic solvent, followed by spectrophotometric determination by Standard Methods 10,150. These analyses were performed in triplicates according to the procedures described by APHA (2017) (Table 1).

2.5 Statistical Analysis

The multivariate statistical analysis was initially performed using a Pearson correlation matrix to assess the relationships among the water-quality variables and to identify those with sufficient intercorrelation to support subsequent multivariate analyses. Pearson's correlation coefficients were considered significant at $p \leq 0.05$, and variables showing absolute correlation coefficients ($|r| \geq 0.55$) were considered relevant for the multivariate approach (Hair Júnior et al. 2014).

Subsequently, principal component analysis (PCA), cluster analysis (CA), and factor analysis (FA) were applied to investigate similarities and dissimilarities among sampling sites and sampling periods. These analyses aimed to identify the main patterns of spatial and temporal variability in water quality and to provide insights into potential natural and anthropogenic sources affecting the studied river reach, based on the

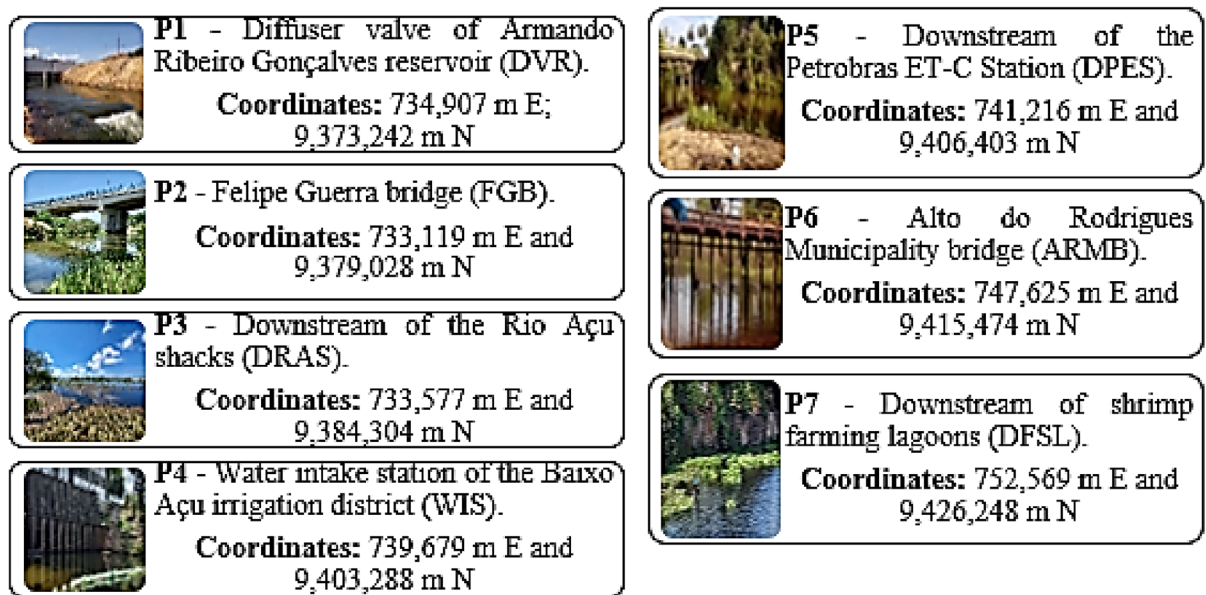


Fig. 5 Collection points and their respective identification codes (SIRGAS 2000/UTM Zone 24S)

variables that contributed most strongly to the differentiation among samples.

For PCA, components were retained according to the Kaiser criterion, considering eigenvalues greater than 1.0 (Kaiser, 1958). Cluster analysis was subsequently used to identify groups of sampling sites and/or sampling periods exhibiting similar water-quality characteristics. Factor analysis was then applied to improve the interpretation of the underlying relationships among variables. The extracted factors were subjected to Varimax orthogonal rotation, which maximizes high factor loadings and minimizes low loadings, thereby facilitating the identification and interpretation of the variables most strongly associated with each factor.

All statistical analyses were performed using R v. 4.6.1 (R Core Team, 2026).

3 Results

3.1 Pearson Correlation Matrix of Physical, Chemical and Microbiological Water Attributes

The Pearson correlation matrix showed negative correlations when evaluating pH with K^+ , HCO_3^- and N ($r=-0.56$; $r=-0.55$ and $r=-0.67$), respectively; K^+ with ALK and P ($r=-0.62$; $r=-0.68$); and

ALK with N ($r=-0.56$). Positive correlations were observed when evaluating pH with ALK and P (0.68; 0.58), in that order; Na^+ with Ca^{2+} , Mg^{2+} and Cl^- ($r=0.85$; $r=0.70$; and $r=0.68$) concomitantly; ALK with P ($r=0.77$); EC with Na^+ and Cl^- ($r=0.59$; and $r=0.60$), respectively; Ca^{2+} with Mg^{2+} ($r=0.56$); Mg^{2+} with Cl^- ($r=0.55$); and P with Chl a ($r=0.60$). Independence was observed for the variables CO_3^{2-} and TC, which showed no correlation with any other attribute (Fig. 7).

3.2 Principal Component Analysis

Principal Component Analysis (PCA) was used as an exploratory approach to summarize the multivariate structure of the water quality dataset. In the $PC1 \times PC2$ plane, the first two principal components accounted for 59.50% of the total variance, with PC1 and PC2 explaining 31.23% and 28.27%, respectively (Fig. 8).

The $PC1 \times PC3$ plane explained 44.50% of the total variance, with PC1 accounting for 31.23% and PC3 for 13.27% (Fig. 9).

3.3 Cluster Analysis

Hierarchical cluster analysis, based on Ward's method and Euclidean distance, at a 30% dissimilarity level,

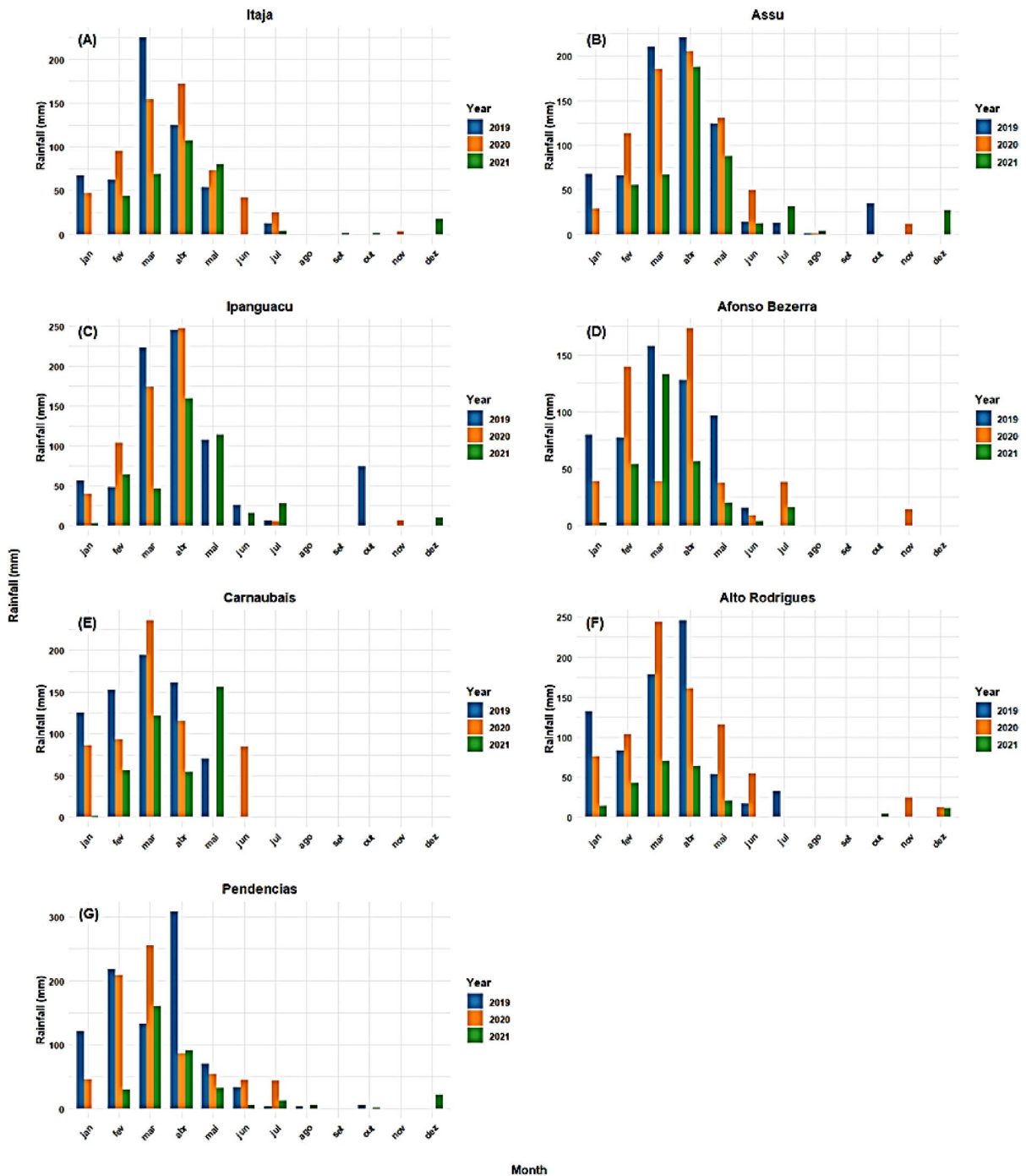


Fig. 6 Monthly rainfall during 2019, 2020 and 2021 in the municipalities of Itajá (A), Assu (B), Ipanguacu (C), Afonso Bezerra (D), Carnaubais (E), Alto do Rodrigues (F) and Pendências (G). Collections C1, C2, and C3 correspond to June,

September, and November 2019, respectively, representing the dry season, whereas C4 and C5 correspond to March 2020 and March 2021, respectively, representing the rainy season

Table 1 Mean values of physicochemical and microbiological water variables obtained from three replicates at different sampling points and collection times

Ident	pH	EC (dS m ⁻¹)	K ⁺ (meq L ⁻¹)	Na ⁺ (meq L ⁻¹)	Ca ²⁺ (meq L ⁻¹)	Mg ²⁺ (meq L ⁻¹)	Cl ⁻ (meq L ⁻¹)	CO ₃ ²⁻ (meq L ⁻¹)	HCO ₃ ⁻ (meq L ⁻¹)	ALK (mg L ⁻¹)	N (mg L ⁻¹)	P (mg L ⁻¹)	Chl a (µg L ⁻¹)	TC MPN (100 mL ⁻¹)	
C1	P1	8.28	0.24	0.16	1.03	0.54	0.86	1.54	0.00	1.73	81.22	0.20	0.50	66.35	114.64
	P2	8.22	0.23	0.16	1.03	0.50	0.80	1.50	0.00	1.80	85.38	0.15	0.65	48.76	114.64
	P3	8.65	0.25	0.16	0.99	0.70	0.60	1.40	0.00	1.70	91.10	0.26	0.49	23.68	114.64
	P4	8.97	0.28	0.16	1.12	0.80	0.60	1.60	0.30	1.70	82.47	0.16	0.21	7.19	114.64
	P5	9.01	0.46	0.19	2.61	0.90	0.70	1.80	0.50	1.60	80.60	0.26	0.21	5.76	98.47
	P6	8.39	0.48	0.18	2.39	1.20	0.90	2.20	0.00	1.90	84.66	0.11	0.29	4.80	98.47
	P7	8.28	0.73	0.20	3.89	1.40	1.10	4.80	0.00	2.10	97.45	0.23	0.31	14.39	98.47
C2	P1	8.65	0.25	0.19	1.25	0.70	0.60	1.40	0.40	2.10	79.14	0.39	0.35	64.75	166.00
	P2	7.92	0.25	0.17	1.21	0.60	0.61	1.20	0.20	1.97	83.72	0.33	0.38	49.16	166.00
	P3	8.34	0.24	0.18	1.34	0.70	0.60	1.20	0.40	1.87	103.90	0.37	0.32	23.53	166.00
	P4	7.93	0.26	0.17	1.34	0.70	0.57	1.40	0.20	2.00	82.37	0.32	0.26	7.49	166.00
	P5	7.67	0.28	0.18	1.49	0.80	0.60	2.06	0.20	1.90	74.98	0.37	0.18	5.52	166.00
	P6	7.75	0.38	0.16	1.96	0.90	0.60	2.20	0.40	1.90	82.68	0.28	0.25	3.84	101.20
	P7	7.54	0.29	0.20	4.75	1.30	1.20	5.20	0.20	2.50	101.61	0.39	0.32	11.51	114.64
C3	P1	7.83	0.25	0.15	1.48	0.70	0.01	2.00	0.20	1.50	83.30	0.58	0.64	36.57	166.00
	P2	8.52	0.25	0.16	1.04	0.70	0.50	2.00	0.40	1.50	83.72	0.52	0.88	49.16	166.00
	P3	8.43	0.24	0.16	0.96	0.60	0.60	2.20	0.40	1.70	78.31	0.49	0.53	18.23	166.00
	P4	8.05	0.25	0.16	1.44	0.60	0.60	2.00	0.20	1.70	82.47	0.47	0.45	7.34	93.56
	P5	7.79	0.33	0.17	1.56	0.60	0.80	2.00	0.20	1.90	86.22	0.48	0.49	1.87	166.00
	P6	7.74	0.32	0.17	1.56	0.90	0.50	1.80	0.20	2.00	86.63	0.46	0.71	4.56	114.64
	P7	7.62	0.44	0.18	2.48	0.90	0.80	2.20	0.20	1.90	93.29	0.56	0.72	5.88	104.32
C4	P1	7.41	0.27	0.18	0.08	0.32	0.51	1.27	0.00	2.13	48.67	0.59	0.03	6.00	166.00
	P2	7.04	0.27	0.18	0.09	0.35	0.53	1.53	0.00	2.17	48.88	0.73	0.01	4.32	114.64
	P3	7.35	0.28	0.18	0.10	0.34	0.47	1.87	0.00	2.07	60.31	0.40	0.02	4.56	104.71
	P4	7.12	0.33	0.18	0.13	0.42	0.60	1.93	0.00	2.50	47.33	0.42	0.01	3.12	93.56
	P5	6.92	0.31	0.19	0.12	0.42	0.44	2.27	0.00	2.37	57.83	0.58	0.01	3.84	77.88
	P6	6.92	0.30	0.19	0.11	0.42	0.44	2.13	0.00	2.17	54.08	0.51	0.01	5.28	87.02
	P7	6.96	0.48	0.24	2.61	0.44	0.82	3.40	0.00	2.27	61.36	0.67	0.02	4.56	3.00

Table 1 (continued)

Ident	pH	EC (dS m ⁻¹)	K ⁺ (meq L ⁻¹)	Na ⁺ (meq L ⁻¹)	Ca ²⁺ (meq L ⁻¹)	Mg ²⁺ (meq L ⁻¹)	Cl ⁻ (meq L ⁻¹)	CO ₃ ²⁻ (meq L ⁻¹)	HCO ₃ ⁻ (meq L ⁻¹)	ALK (mg L ⁻¹)	N (mg L ⁻¹)	P (mg L ⁻¹)	Chl a (µg L ⁻¹)	TC MPN (100 mL ⁻¹)
C5	P1	7.31	0.25	0.22	1.28	0.66	0.58	1.00	0.30	1.60	0.60	0.01	14.87	166.00
	P2	7.44	0.24	0.22	1.20	0.61	0.65	1.10	0.10	1.65	0.94	0.02	5.28	101.20
	P3	7.85	0.23	0.22	1.24	0.74	0.58	1.00	0.40	1.65	0.31	0.03	1.92	114.64
	P4	7.44	0.25	0.22	1.38	0.80	0.66	1.30	0.35	1.85	0.37	0.02	2.40	101.20
	P5	6.96	0.26	0.23	1.51	0.78	0.66	1.20	0.30	1.90	0.67	0.04	2.40	87.02
	P6	7.19	0.31	0.23	1.68	0.91	0.71	1.00	0.25	1.95	0.56	0.04	5.28	93.56
	P7	7.02	0.35	0.24	2.07	1.02	0.69	1.10	0.30	2.10	0.75	0.03	0.96	101.20

EC, electrical conductivity; ALK, alkalinity; Chl a, chlorophyll a; TC, thermotolerant coliforms; P, points; C, collections; C1, C2, and C3 correspond to samples collected in June, September, and November 2019, respectively, during the dry season. C4 and C5 correspond to samples collected in March 2020 and March 2021, respectively, during the rainy season; P1 – Diffuser valve of Armando Ribeiro Gonçalves Reservoir; P2 – Felipe Guerra Bridge; P3 – Downstream of the Rio Açu shacks; P4 – Water intake station of the Baixo Açu Irrigation District; P5 – Downstream of the Petrobras ET-C Station; P6 – Alto do Rodrigues Municipality Bridge; and P7 – Downstream of shrimp-farming lagoons

separated the 35 samples into three distinct groups representing multivariate water quality clusters for the studied water source (Fig. 10).

3.4 Factor Analysis

Factor analysis (Table 2) was employed to identify sets of physicochemical and microbiological variables exhibiting similar variation patterns across water samples, complementing the interpretation derived from Principal Component Analysis (Figs. 8 and 9) and Hierarchical Cluster Analysis (Fig. 10). Of the 14 variables evaluated, 13 showed high factor loadings ($|\text{loading}| \geq 0.60$) on at least one of the retained factors. Three factors (F1—Trophic–hydrochemical factor, F2—Mineralization–salinity factor, and F3—Carbonate–bicarbonate equilibrium factor) were identified, all with eigenvalues greater than 1.0, which collectively explained 71.93% of the total variance in the dataset.

4 Discussion

The results indicate that both pressures associated with land use and land cover and hydroclimatic seasonality contributed to the observed variability in the hydrochemical, trophic, and microbiological conditions of the water, thereby corroborating the hypothesis proposed for the study.

Comparing the mean values of the 21 samples from the dry period (C1, C2, and C3) with the 14 samples from the rainy period (C4 and C5) reveals a marked seasonal response across the 14 water quality variables (Table 1). Based on rainfall distribution in the study area municipalities (Figs. 1 and 6), C4 and C5 coincided with the period of highest precipitation, whereas C1, C2, and C3 occurred predominantly during the dry season. During the rainy period, the greatest reductions were observed in P (95.10%) and chlorophyll-a (78.90%), indicating a reduced phytoplankton response. Levels of Na⁺ (44.70%), ALK (41.0%), CO₃²⁻ (34.80%), Ca²⁺ (26.30%), Cl⁻ (24.10%), thermotolerant coliforms (0.12 log), Mg²⁺ (15.20%), and EC (−7.50%) also decreased, while pH dropped from 8.17 to 7.21, reflecting less alkaline conditions. Conversely, N (64.60%), K⁺ (21.30%), and HCO₃⁻ (9.20%) increased, suggesting that rainfall promoted not only dilution but also the mobilization

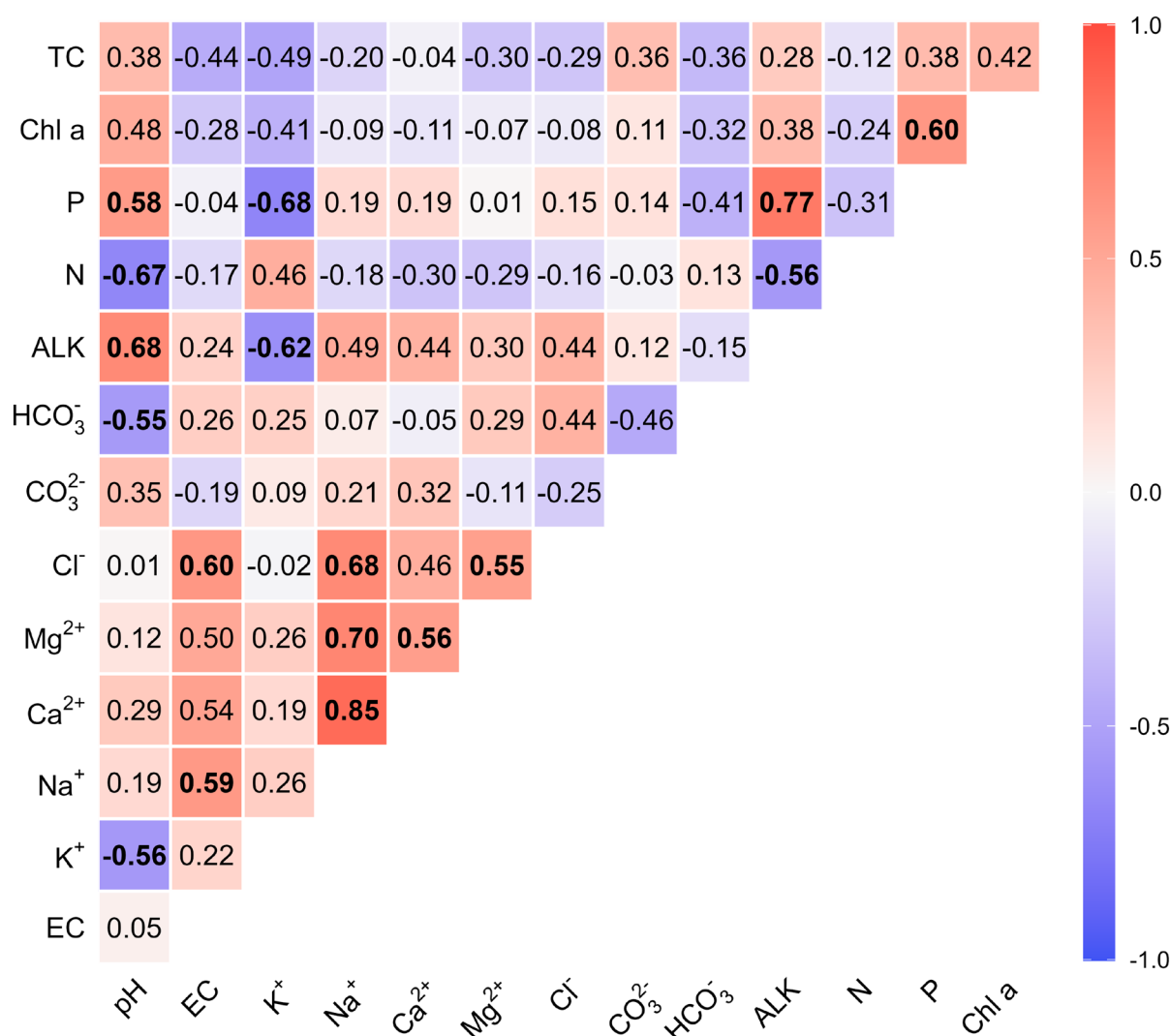


Fig. 7 Heatmap of the Pearson correlation matrix for water quality variables (physical, chemical, and microbiological), based on averages from all sampling points and collection periods. Note: EC – electrical conductivity; ALK – alkalinity;

Chl a – chlorophyll a; TC – thermotolerant coliforms. $n=35$, $p \leq 0.05$, and coeficiente de correlação de Pearson (r) showing absolute correlation coefficients ($|r| \geq 0.55$)

and selective transport of constituents from the drainage basin.

This behavior is consistent with the mosaic of land uses and occupancy observed in Fig. 4, which includes agricultural areas, pastures, temporary and perennial crops, urbanized areas, and aquaculture along the studied reach. The specific location of the sampling points (Fig. 5) reinforces this heterogeneity: P1 represents water released from the Armando Ribeiro Gonçalves reservoir; P2 and P3 are more subject to human activity; P4 is associated with the Baixo

Açu Irrigation District; P5 is situated downstream of the Petrobras ET-C station; and P7 is located downstream of shrimp farming ponds. Thus, the observed patterns suggest that hydroclimatic seasonality simultaneously influences water dilution and turnover, as well as the transport of solutes and nutrients derived from various land uses, explaining the simultaneous decrease in ions and P and the increase in N, K⁺, and HCO₃⁻ during the rainy season.

Water quality assessment for various uses revealed distinct responses among the analyzed parameters

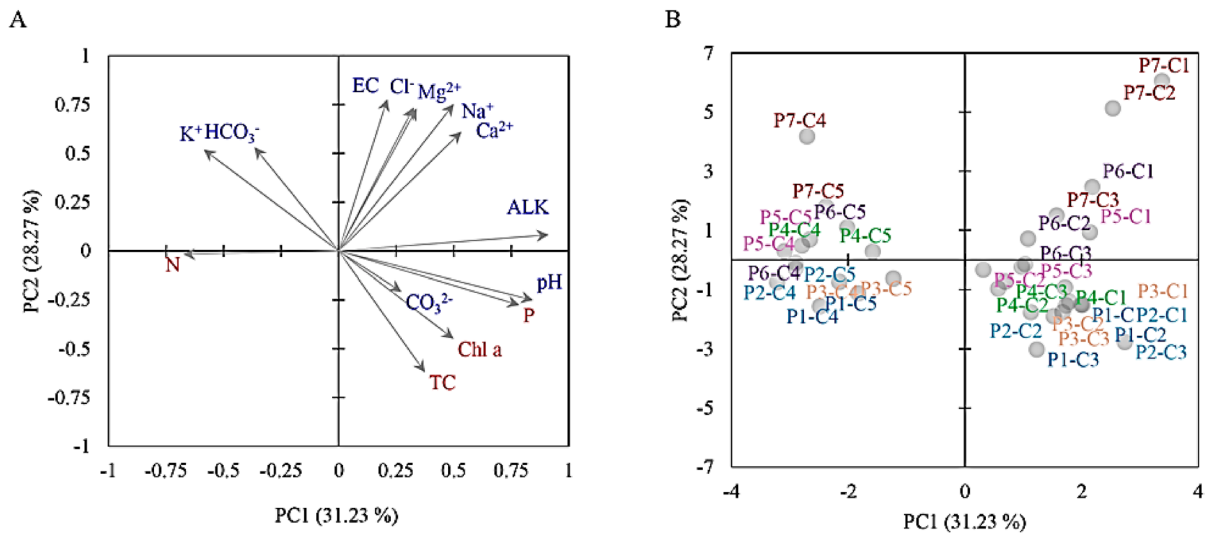


Fig. 8 Principal component analysis of water-quality variables: **A** correlation circle showing the relationships between variables and the first two principal components (PC1 and PC2); and **(B)** PCA score plot showing the distribution of sampling point-collection combinations in the PC1 × PC2 plane. Note: EC – electrical conductivity; ALK – alkalinity; Chl a – chlorophyll a; TC – thermotolerant coliforms; PC – principal components; P – points; C – collections; C1, C2, and C3 correspond to samples collected in June, September, and

November 2019, respectively, during the dry season. C4 and C5 correspond to samples collected in March 2020 and March 2021, respectively, during the rainy season; P1 – Diffuser valve of Armando Ribeiro Gonçalves Reservoir; P2 – Felipe Guerra Bridge; P3 – Downstream of the Rio Açu shacks; P4 – Water intake station of the Baixo Açu Irrigation District; P5 – Downstream of the Petrobras ET-C Station; P6 – Alto do Rodrigues Municipality Bridge; and P7 – Downstream of shrimp-farming lagoons

(Table 1). Regarding guidelines for human consumption, only 28.57% of the evaluated variables had directly comparable reference values (WHO 2026). During the dry season, 28.57% of the sampling points showed pH > 8.40, whereas Na⁺ and Cl⁻ levels remained below their respective organoleptic thresholds of 8.70 and 7.05 meq L⁻¹ in all samples. However, thermotolerant coliforms were detected at all points, failing to meet the criterion of absence required for direct consumption.

For irrigation purposes, 82.86% of the points showed pH values within the standard range of 6.5–8.4, and 97% exhibited EC < 0.70 dS m⁻¹, indicating no salinity-related restrictions (Ayers & Westcot, 1985). Furthermore, 94.29% of the samples showed Na⁺ < 3.0 and Cl⁻ < 4.0 meq L⁻¹, while 100% remained within the standard ranges for Ca²⁺ (0–20 meq L⁻¹) and Mg²⁺ (0–5 meq L⁻¹). Conversely, K⁺ levels exceeded the standard range in all samples; 62.86% showed CO₃²⁻ > 0.1 meq L⁻¹, and all samples fell within the slight-to-moderate restriction range for HCO₃⁻ (1.5–8.5 meq L⁻¹). Regarding the potential for emitter clogging, 34.29% of the points

presented a severe risk associated with pH (> 8.0), whereas EC, Ca²⁺, and Mg²⁺ levels indicated low risk across all samples (Bucks et al., 1979; Capra & Scicolone, 1998). For aquaculture, 97.14% of the sampling points showed pH levels between 6.0 and 9.0, and 100% showed alkalinity between 20 and 300 mg L⁻¹, a range considered suitable for fish development (FAO 2003), although more specific requirements may vary depending on the cultured organisms (Wurts, 2002). Direct comparisons between the results of the present study and those reported by previous authors should be made with caution, as the sampling sites are located at different geographic coordinates and are therefore subject to distinct land-use and land-cover conditions, which may differentially affect water-quality patterns (Câmara et al., 2015; Chellappa et al., 2009; Costa et al., 2006; Vieira et al., 2015).

The Pearson correlation heatmap (Fig. 7) revealed that 18.68% of the correlations between variables were significant, with all of them classified as strong (Turney 2024). pH exhibited the highest number of significant associations (29.41%), highlighting its

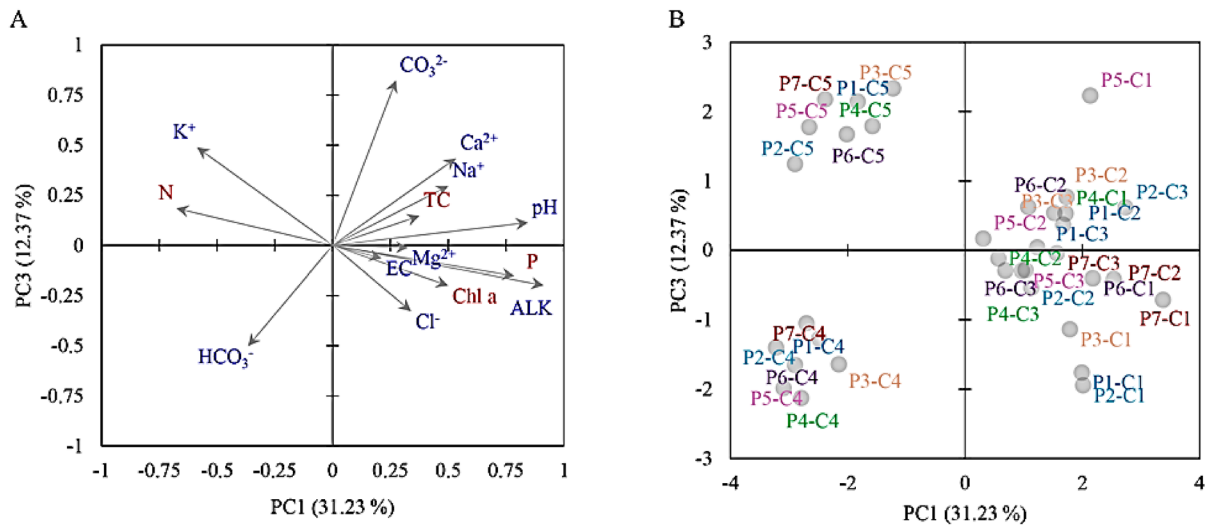


Fig. 9 Principal component analysis of water-quality variables: **A** correlation circle showing the relationships between variables and the first two principal components (PC3 and PC1); and **(B)** PCA score plot showing the distribution of sampling point-collection combinations in the PC3×PC1 plane. Note: EC – electrical conductivity; ALK – alkalinity; Chl a – chlorophyll a; TC – thermotolerant coliforms; PC—principal components; P – points; C – collections; C1, C2, and C3 correspond to samples collected in June, September, and

November 2019, respectively, during the dry season. C4 and C5 correspond to samples collected in March 2020 and March 2021, respectively, during the rainy season; P1 – Diffuser valve of Armando Ribeiro Gonçalves Reservoir; P2 – Felipe Guerra Bridge; P3 – Downstream of the Rio Açu shacks; P4 – Water intake station of the Baixo Açu Irrigation District; P5 – Downstream of the Petrobras ET-C Station; P6 – Alto do Rodrigues Municipality Bridge; and P7 – Downstream of shrimp-farming lagoons

sensitivity to biological processes, water–soil–rock interactions, and rainfall seasonality. The negative correlation with K^+ may be linked to its mobilization via surface runoff and its release during the decomposition and leaching of organic matter (Chaudhuri et al., 2007; Oliveira Júnior et al., 2021), whereas the inverse relationship with HCO_3^- may be associated with carbonate system equilibrium, given that the photosynthetic consumption of CO_2 and H^+ promotes a rise in pH and shifts the equilibrium from HCO_3^- toward CO_3^{2-} (Sand-Jensen et al., 2019). Meanwhile, the positive association between pH and alkalinity suggests a geogenic contribution and higher water buffering capacity, intensified by the mobilization of alkaline species during the rainy season (Lauerwald et al., 2013). The negative relationship between pH and N may be linked to the transformation of nitrogenous compounds, specifically nitrification, which releases H^+ . Conversely, the positive correlations between pH and P, and between P and chl a, suggest the influence of eutrophication, as increased phytoplankton production intensifies CO_2 consumption, thereby promoting a rise in pH (Chellappa et al.,

2009; Cortez et al., 2022; Costa et al., 2006; da Silva et al., 2026; Lopes et al., 2014; Vieira et al., 2015).

Furthermore, the positive correlations of EC with Na^+ and Cl^- , as well as of Na^+ with Ca^{2+} , Mg^{2+} , and Cl^- , indicate the contribution of these ions to water salinity. This behavior may result from the interaction between the weathering and dissolution of minerals from basin soils and rocks, salt concentration via evaporation during dry periods, and irrigation return flows, which can transport salts accumulated in the soil into water bodies (Fontes et al., 2015; Jafari et al., 2012; Monteiro et al., 2022). The association between Ca^{2+} and Mg^{2+} reinforces a possible common geogenic origin linked to mineral weathering and dissolution (Sun et al., 2014), whereas the relationship between Mg^{2+} and Cl^- highlights salinization and evapoconcentration processes typical of semi-arid environments (Monteiro et al., 2022). In contrast, the negative correlations of K^+ with ALK and P suggest distinct dynamics for this ion, the mobilization of which from agricultural soils may be facilitated by runoff during rainfall events (Sharpley et al., 1988). The inverse relationship between ALK

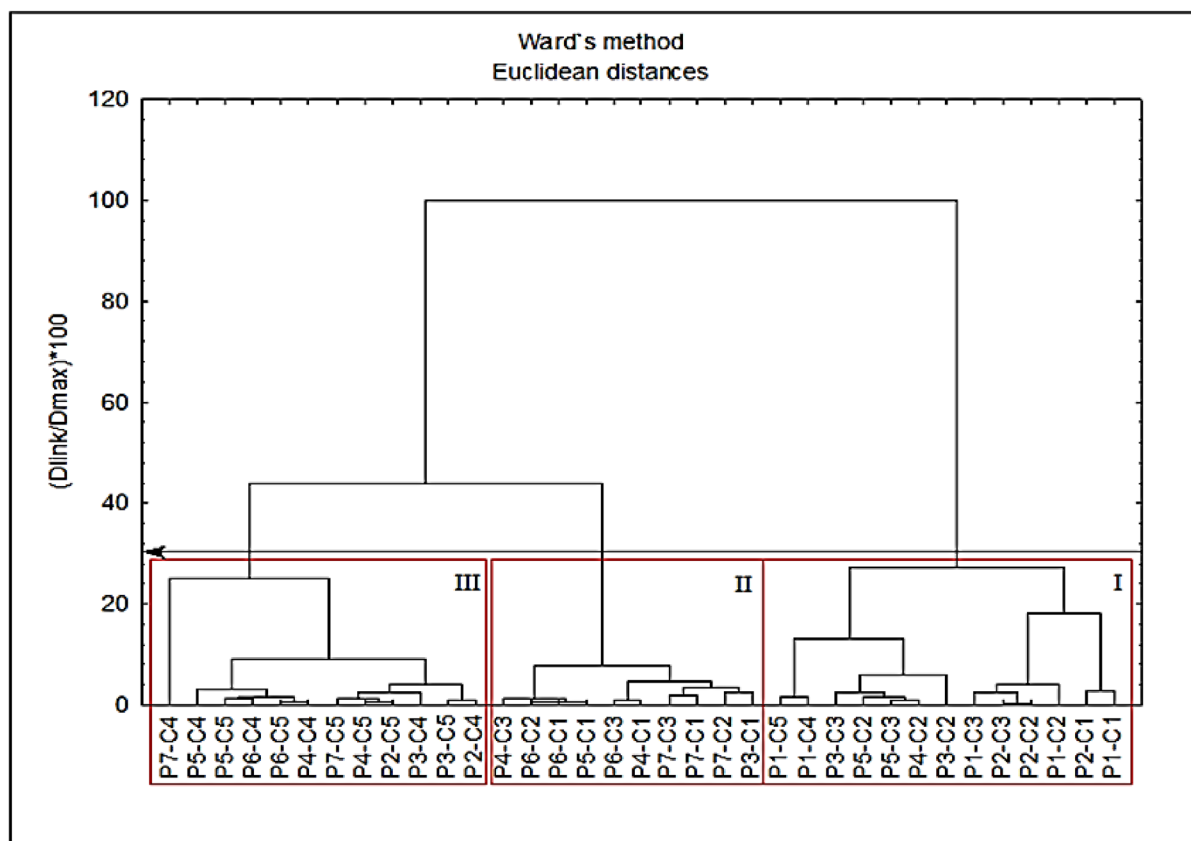


Fig. 10 Euclidean distance and group formation for the different collection points and seasons. Note: EC – electrical conductivity; ALK – alkalinity; Chl a – chlorophyll a; TC – thermotolerant coliforms; P – points; C – collections; C1, C2, and C3 correspond to samples collected in June, September, and November 2019, respectively, during the dry season. C4 and C5 correspond to samples collected in March 2020 and March

2021, respectively, during the rainy season; P1 – Diffuser valve of Armando Ribeiro Gonçalves Reservoir; P2 – Felipe Guerra Bridge; P3 – Downstream of the Rio Açu shacks; P4 – Water intake station of the Baixo Açu Irrigation District; P5 – Downstream of the Petrobras ET-C Station; P6 – Alto do Rodrigues Municipality Bridge; and P7 – Downstream of shrimp-farming lagoons

and N may be linked to nitrification, a process that produces H^+ and consumes alkalinity (Singh et al., 2022), while the positive association between ALK and P may reflect the interplay between hydrochemical conditions, basin inputs, and processes related to eutrophication and phosphorus cycling (Spears et al., 2013). Finally, the lack of significant correlations between TC and the other variables may be related to the complexity of its dynamics in aquatic environments, influenced jointly by biotic, abiotic, and physicochemical factors (Reche et al., 2010), a behavior also observed by da Silva et al. (2026).

Regarding the principal component analysis, PC1 was primarily associated with a gradient characterized by positive relationships with ALK, pH, P, Chl

a, TC, and CO_3^{2-} , whereas K^+ , HCO_3^- , and N were positioned in the opposite direction. PC2, in turn, primarily represented salinity and ionic composition, with EC, Cl^- , Mg^{2+} , Na^+ , and Ca^{2+} showing the strongest positive associations, while TC, Chl a, P, and CO_3^{2-} were associated with negative scores (Fig. 8A). The score plot further revealed a clear seasonal pattern: samples from C4 and C5 were predominantly positioned on the negative side of PC1, whereas several samples from C1, C2, and C3 showed positive scores (Fig. 8B). Samples located close to one another exhibited similar physicochemical and biological characteristics, while greater distances reflected increasing differences in multivariate composition.

Table 2 Factor loadings extracted from the physico-chemical and microbiological water variables for the different collection points and times and their respective factor loadings, eigenvalues, total variance (%) and accumulated variance (%)

Variables	Factorial loads		
	Factor 1	Factor 2	Factor 3
pH	0.77	0.18	0.40
EC	−0.07	0.76	−0.26
K ⁺	−0.88	0.29	0.14
Na ⁺	0.05	0.94	0.16
Ca ²⁺	0.08	0.85	0.33
Mg ²⁺	0.01	0.79	−0.16
Cl [−]	0.16	0.73	−0.45
CO ₃ ^{2−}	0.01	0.09	0.89
HCO ₃ [−]	−0.32	0.21	−0.72
ALK	0.83	0.44	0.03
N	−0.63	−0.29	0.00
P	0.84	0.08	0.15
Chl a	0.65	−0.21	0.16
TC	0.49	−0.35	0.43
Eigenvalue	4.39	3.97	1.72
Total variance (%)	31.32	28.36	12.25
Cumulative (%)	31.32	59.68	71.93

EC, electrical conductivity; ALK, alkalinity; Chl a, chlorophyll a; TC, thermotolerant coliforms; Factor loadings with absolute values ≥ 0.60 are shown in bold. Factor 1—Trophic–hydrochemical factor; Factor 2—Mineralization–salinity factor; Factor 3—Carbonate–bicarbonate equilibrium factor

This seasonal separation was consistent with the higher rainfall recorded during C4 and C5 (Fig. 6), which likely increased water volume and turnover, favoring the dilution of salts, nutrients, and phytoplankton biomass, a pattern commonly reported for semi-arid reservoirs (Chellappa et al., 2009; Costa et al., 2006; Vieira et al., 2015). Consequently, ALK, P, and Chl a decreased during C4 and C5 (Table 1). However, heavy rainfall can also promote initial nutrient pulses and water column mixing, as previously observed in the Armando Ribeiro Gonçalves Reservoir (Câmara et al., 2015), indicating that the hydrochemical response depends on rainfall intensity and the stage of the rainy season. Spatially, sites P5, P6, and P7 were more strongly associated with salinity-related variables, indicating increasing ionic enrichment in a downstream direction. This pattern may reflect salt concentration during the dry season due to intense evaporation, combined with inputs

from irrigated agriculture, irrigation return flows, and shrimp farming. In particular, site P7, located downstream of shrimp farms, showed a strong association with EC, Na⁺, Cl[−], Ca²⁺, and Mg²⁺, recording the highest Na⁺ and Cl[−] concentrations during the dry season; this suggests a local contribution to salinization, given that shrimp farming effluents can increase salt loading into receiving water bodies (de Paiva et al., 2024). Conversely, sites P1 through P4 showed a stronger association with P, Chlorophyll-a (Chl a), and TC, suggesting a more pronounced influence of nutrient enrichment and microbiological contamination. Such conditions may be linked to fertilizer transport via surface runoff from agricultural areas (Baldassarini & Nunes, 2018; Manaouch et al., 2020) and the input of untreated domestic wastewater (Baretta et al., 2019). The higher Chl a concentrations observed at P1 and P2 during the dry season are also consistent with the increased trophic levels reported in semi-arid region reservoirs under low water storage conditions, as well as with the known seasonal phytoplankton variability in the Armando Ribeiro Gonçalves Reservoir (Chellappa et al., 2009; Wiegand et al., 2021). Similarly, point P4, located within the Baixo Açu Irrigated Perimeter, may receive inputs from irrigated agriculture, given that evapotranspiration concentrates salts in the soil and irrigation return flow can transport these solutes, particularly Cl[−], into adjacent water bodies (Jafari et al., 2012).

In addition to seasonality and land-use pressures, hydrochemical variability may also reflect the geological and pedological heterogeneity of the study area (Figs. 2 and 3). According to approximate correspondence with USDA Soil Taxonomy, the basin is dominated by Entisols, particularly Fluvents associated with fluvial environments, with additional occurrences of Inceptisols, Oxisols, and shallow Entisols related to lithic substrates. These soils occur over a heterogeneous geological framework comprising alluvial and colluvial deposits, coastal-tableland materials, interbedded silt–clay and quartz-rich sandy sediments, undifferentiated migmatites, and quartz-rich sandy and conglomeratic deposits. Less developed soils, particularly Entisols and Inceptisols, may contain minerals more susceptible to weathering and may contribute Ca²⁺, Mg²⁺, Na⁺, K⁺, and HCO₃[−] through water–soil and water–sediment interactions. In contrast, highly weathered Oxisols are generally dominated by low-activity clays and Fe and Al oxides,

favoring P adsorption and retention. Thus, spatial differences in ionic composition and nutrient availability may partly reflect the combined influence of substrate mineralogy, soil development, and hydrological interactions with the surrounding landscape (Campos et al., 2016; CPRM, 2005a, 2005b, 2005c, 2005d, 2005e; Santos et al., 2025).

The PC1×PC3 analysis provided complementary information regarding the processes controlling water quality (Fig. 9). While PC1 maintained the previously identified seasonal hydrochemical–trophic gradient, PC3 (12.37%) primarily represented variations in the inorganic carbon system. CO_3^{2-} showed the strongest positive association with PC3, followed by K^+ , whereas HCO_3^- was strongly related to negative scores; Chl a, Cl^- , and ALK were also positioned in the negative direction. This configuration indicates that PC3 mainly captures shifts in the carbonate–bicarbonate equilibrium and associated hydrochemical conditions.

The score plot reinforced these differences, with samples of similar multivariate composition positioned closer together and more distinct samples located farther apart (Fig. 9B). Consistent with the PC1×PC2 pattern, C4 and C5 were predominantly associated with negative PC1 scores, whereas most C1, C2 e C3 samples remained on the positive side. More importantly, PC3 differentiated the two rainy-season campaigns: several C4 samples clustered at negative scores and were strongly associated with HCO_3^- , suggesting enhanced bicarbonate availability related to atmospheric CO_2 dissolution, organic matter decomposition, and silicate weathering (Lima et al., 2017). In contrast, CO_3^{2-} was more closely associated with C1, during the dry period, possibly reflecting carbonate-system shifts under conditions of high evaporation and greater concentration of dissolved constituents. Thus, PC3 highlights a distinct seasonal reorganization of inorganic carbon species, superimposed on the broader hydrochemical pattern captured by PC1.

A joint analysis of Figs. 4 and 10 reveals the following patterns: Group I showed the lowest dependence on seasonality, as it included all five samples from P1 and, predominantly, samples from P2 to P5 collected during C1, C2, and C3. This group represented a hydrochemical condition characterized by low salinity, greater eutrophic expression, and higher levels of thermotolerant coliforms. Group II showed

a predominant association with the dry season and consisted mainly of samples from P3 to P7 collected during C1, C2, and C3. This group represented a transitional hydrochemical condition, characterized by intermediate levels of eutrophication and microbiological contamination. In contrast, Group III showed the strongest and clearest association with the rainy season, as it was composed exclusively of samples collected during C4 and C5 from points P2 to P7. This group was characterized by a distinct hydrochemical condition, with no evidence of increased salinity and with lower eutrophic expression and, in general, lower microbiological contamination.

Finally, Factor 1 (F1) explained 31.32% of the total variance (Table 2) and showed high factor loadings for pH (0.77), K^+ (−0.88), alkalinity (0.83), N (−0.63), P (0.84), and chlorophyll-a (0.65). The opposing factor loadings of N and K^+ relative to pH, alkalinity, P, and chlorophyll-a indicate a trophic–hydrochemical gradient characterized by contrasting nutrient and water chemistry conditions among samples, consistent with the seasonal changes observed between the dry and rainy season sampling events (Table 1). Factor 2 (F2) explained 28.36% of the total variance and was characterized by high loadings for EC (0.76), Na^+ (0.94), Ca^{2+} (0.85), Mg^{2+} (0.79), and Cl^- (0.73), reflecting a hydrochemical condition associated with increased dissolved ion content in the water (Freire, 2022). Factor 3 (F3) accounted for an additional 12.25% of the variance and was defined primarily by CO_3^{2-} (0.89) and HCO_3^- (−0.72), representing variations in carbonate and bicarbonate anions. Thermotolerant coliforms did not exhibit high loadings on any of the retained factors, indicating that their variability was independent of the physicochemical structure.

5 Conclusions

The results confirmed that land use and land cover downstream of the Armando Ribeiro Gonçalves Dam are associated with spatial variations in the physical, chemical, and microbiological quality of the Piranhas-Açu River. Areas influenced by irrigated agriculture and shrimp farming exhibited greater ionic enrichment, particularly regarding electrical conductivity and concentrations of Na^+ , Ca^{2+} , Mg^{2+} , and Cl^- ,

indicating a salinization process that intensifies in the downstream reaches. In contrast, upstream and intermediate sites showed a stronger association with P, chlorophyll-a, and thermotolerant coliforms, reflecting nutrient enrichment and microbiological pressure.

Seasonality was also a determining factor in water quality variability. Samples collected during the dry season were characterized by higher levels of salinity, alkalinity, P, and chlorophyll-a, whereas wet-season samples showed lower values for these parameters and a predominance of K^+ , HCO_3^- , and N; this demonstrates that rainfall altered both solute concentrations and hydrochemical dynamics. Principal component analysis made it possible to distinguish gradients related to salinity and ionic composition, trophic and hydrochemical conditions, and the CO_3^{2-} – HCO_3^- equilibrium. Consistently, hierarchical cluster analysis formed three groups with approximately 30% dissimilarity, reinforcing climatic seasonality as a primary factor differentiating sampling conditions.

Overall, the multivariate approach effectively integrated spatial and temporal water quality responses with land use, rainfall seasonality, and the basin's natural geochemical characteristics. Irrigation, shrimp farming, agricultural areas, and urban development were linked to changes in indicators for salts, nutrients, and microbiological parameters, highlighting the potential for progressive salinization and eutrophication processes in the downstream reaches. These results corroborate the need for periodic water quality monitoring and environmental management actions addressing the main anthropogenic pressures identified along the Piranhas-Açu River in Rio Grande do Norte, Brazil.

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Author Contributions Maria Elidayane da Cunha: Conducted fieldwork and laboratory analyses, performed formal data analysis, wrote the original draft, and participated in manuscript review and editing. Mateus Pessoa Caminha: Conducted field experiments and participated in manuscript review and editing. Luis César de Aquino Lemos Filho (Advisor): Provided supervision, developed methodology, performed validation, contributed to manuscript review and editing, and acquired funding. João Everthon da Silva Ribeiro: Performed validation, improved tables and figures visualization, and participated in manuscript review and editing. Rutilene Rodrigues da Cunha: Conducted field experiments and laboratory analyses. Ana

Elidary da Cunha: Conducted field experiments and laboratory analyses. Lunara Gleika da Silva Rego: Conducted field experiments and laboratory analyses. Rafael Oliveira Batista: Contributed to the development of the research and participated in the review and editing of the manuscript. Paulo Cesar Moura da Silva: Prepared the thematic maps and participated in the review and editing of the manuscript. Jeane Cruz Portela: Contributed to the development of the research and participated in the review and editing of the manuscript. Antônio Gustavo de Luna Souto: Performed validation, improved tables and figures visualization, and participated in manuscript review and editing. Luiz Fernando de Sousa Antunes: Performed validation, improved tables and figures visualization, and participated in manuscript review and editing.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Ethical Approval Not applicable.

Consent to Participate Not applicable.

Consent to Publish Not applicable.

Competing interests The authors declare that they have no conflict of interest.

Clinical Trial Number Not applicable.

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